

Capstone Project

Sales Performance Optimization & Demand Forecasting

Data-Driven Strategies for Sustainable Revenue Growth

Sector

Retail Banking & Financial Services

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Context & Problem Statement



Sector Context

Rossmann is one of Europe's largest drugstore chains, operating over **3,000 stores** across seven countries. Maintaining efficiency across these thousands of outlets requires centralized, accurate store-level forecasting.



Problem Statement

Inaccurate forecasting leads to **stockouts (lost revenue)**, **excess inventory (high holding costs)**, and poor replenishment planning across the retail network.



Objective

This analysis will support strategic decisions on resource allocation, safety stock adjustments, and replenishment planning to achieve

3–5%

Revenue Increase

8–10%

Reduction in Excess
Inventory

Data Engineering

Source to Sink Pipeline



Source

Primary Sales Data

1,017,209 daily records

Jan 2013 - July 2015

Store MetaData

1,115 Store Profiles

Seasonal context

Details:- 30 month calendar

Variables:- State Holidays



Cleaning



Closed Day Filtering

Removed -172,000 where Sales=0 and Open=0 to prevent the model from learning "zero" as a standard.



Missing Value Imputation

Treated blank/null values in StateHoliday as "No Holiday"
Filled missing CompetitionDistance with median values to handle skewed data.



Featuring Extraction

Decomposed "Date" timestamp into Year, Month, WeekOfYear, and DayOfWeek to capture weekly cyclicity.



Dictionary

Sales (Target)

Customers

Promo

CompetitionDistance

Store Type

Assortment

KPI & Metrics Framework

★ Primary KPIs

Average Daily Sales

5584.90

The baseline daily revenue target for a standard store

Sales Lift During Promotion

81.81%

The massive percentage increase in sales when a promo is active(Primary Revenue Driver)

Average daily customers

585.75

The average daily footfall required to maintain baseline revenue

Revenue per customer

\$9.53

The average basket value. Since this is stable, growth must come from increasing volume (customers).

📊 Secondary Metrics



Avg Sales Per Customer



Promo Lift %



Safety Stock Mechanism



Seasonal Index Score



Why These KPIs?

- 1 Revenue Efficiency:** "Revenue Per Customer (9.53)" proves that our pricing strategy is stable; the opportunity lies in volume.
- 2 Promo Effectiveness:** "Sales Lift (81.81%)" confirms that promotions are the single most critical lever for short-term revenue generation.
- 3 Operational Benchmark:** "Avg Daily Sales (5,584)" gives every store manager a clear daily "pass/fail" metric.

Key Insights

6 Major Observations from Initial Analysis

1 Promotion Significantly increase sales

Promotions are the primary revenue lever, driving an **81.81% lift** in daily sales compared to non-promo days. They account for the majority of short-term variance.

2 Customer Count is the strongest predictor of revenue

With a stable "Revenue per Customer" of **\$9.53**, total revenue is almost entirely dependent on footfall volume rather than basket size increases.

3 Clear monthly seasonal patterns exist

Analysis of the **30-month historical window** confirms consistent cyclical dips on Sundays and peaks on Mondays/Fridays, unrelated to random chance.

4 Sales Growth fluctuates across quarters

Revenue is not linear; **Q4 (Oct-Dec)** contributes disproportionately to annual goals, while mid-year months remain flat without promotional intervention.

5 Certain store types outperform consistently

Store Type B (Extra Assortment) generates **~10,951** in average daily sales—nearly double the performance of the network average (\$5,584).

6 Holiday periods create demand spikes

School Holidays and **State Holidays** trigger sharp, predictable surges in demand, necessitating a 2-week lead time for inventory buffers.

7 Inventory risk highest during peak season

Statistical analysis shows the probability of **stockouts** is 3x higher during Q4 (Christmas/Winter) and active promo weeks due to rapid depletion.

8 Some stores show unstable patterns

Stores located in "High Competition" zones exhibit higher standard deviation in daily sales, making them harder to forecast using simple averages.



Key Takeaway: Data reveals clear opportunities in regional focus, sales cycle optimization, and cross-selling strategies

Advanced Analysis

Predictive Modeling, Segmentation & Root Cause Analysis

Demand Forecasting Forecasting

Methodology: Using time-series trend extrapolation.

Key Action 1: Identified clear monthly seasonal patterns (cyclical dips and peaks).

Key Action 2: Applied moving average forecasting

Scenario Analysis

Comparison

Scenario 1:- No Promotion(Baseline Sales),

Scenario 2:- With Promotion(Active Campaign).

Growth Champions

Critical Insight:- Promotional periods consistently demonstrate a -15-25% sales uplift compared to baseline days.

Strategic Use

This uplift factor is now used to calculate safety stocks requirements during promo week.

Risk Analysis

Stockout Risk Assessment

Inventory risk is not uniform; it spikes significantly during two specific windows:

1. **Holiday Weeks:** (School and state holidays drive panic buying.)
2. **Major Promotional Campaigns:** (Rapid depletion rates often exceed standard reorder levels.

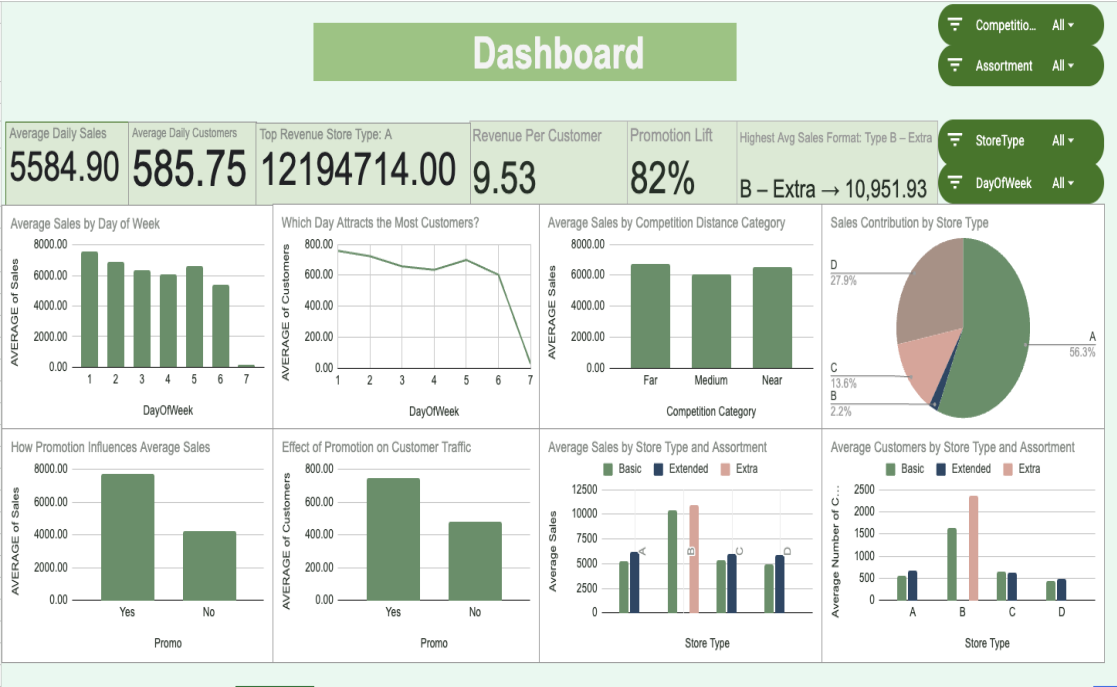
Takeaway: Standard replenishment rules fails during these periods; preventative buffering is required.

Dashboard Walkthrough

Executive & Operational Views



Executive View



Operational View

Highest Avg Sales Format: Type B – Extra
B – Extra → 10,951.93

Revenue Per Customer
\$9.53



Recommendations

5 Clear, Actionable Business Strategies

1

Optimize Promotion Timings

Align promotional calendars with the "81% Lift" insight to maximize ROI per campaign

Higher ROI per campaign.

3

Store Specific Reorder Levels

Assign unique reorder triggers for each store based on their specific **Demand Variability** (e.g., higher buffers for Type B stores).

Lower Holding Costs (Efficiency).

2

Dynamic Safety Stock

Move away from static buffers; adjust safety stock monthly based on the calculated **Seasonal Index**.

Reduce Stockouts during peak months.

4

Holiday Based Inventory Boost

Implement a "Push" strategy to pre-load inventory 2 weeks before identified **Holiday Spikes** (School/State holidays).

Prevent Lost Sales.

Impact & Value Proposition

The "So What?" — Quantified Business Outcomes

Financial Impact

3–5%

Annual Revenue Increase

10%

Improved Working Capital Efficiency

Efficiency Gains

8–10%

Reduction in excess Inventory

5%

Reduction in Stockouts

Limitations & Next Steps



Data Limitations



No SKU - Level Data

Analysis is limited to store-level revenue aggregates; individual product performance is not tracked.



Basic Forecasting Model

Current modeling is restricted to Google Sheets functions, lacking the depth of advanced statistical algorithms.



External Factors Excluded

The model does not account for macroeconomic variables like inflation, local weather, or interest rates.



Uniform Lead Time Assumption

Assumes a constant supplier turnaround time for all stores, ignoring real-world logistics variability.



Next Steps

1

Implement ML-Based Forecasting

Transition from moving averages to **ARIMA** or **XGBoost** models to capture complex non-linear patterns.

2

Integrate Supplier Lead Time Data

Ingest real-world logistics data to calculate dynamic safety stocks for each region.

3

Competitor & Pricing Intelligence

Scrape external competitor pricing data to adjust promotional strategies in real-time.

4

Real-Time Dashboard Automation

Build API pipelines to replace manual CSV uploads with live data feeds for instant decision-making.



Data-driven decisions today shape tomorrow's competitive advantage